

**A Project Report**  
**on**  
**Building a QA Chatbot on**  
**Ancient Greece Using Advanced AI**  
**Frameworks**

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**OBJECTIVE:** The main objective is to develop a robust and accurate QA (Question-Answering) chatbot capable of answering questions about Ancient Greece using a provided dataset. The chatbot must minimize hallucinations and ensure all answers are directly supported by the data.

## 1. WORKFLOW

The detailed steps for the proposed project are illustrated in **Figure. 1** and explained below:

1. **Load the Dataset:** Load the dataset consisting of 60 individual text files.
2. **Text Preprocessing:** Apply preprocessing techniques to clean the text data by removing special characters, noise, stop words, etc.
3. **Determine Optimal Chunking Parameters:** Use a novel chunking technique to determine the optimal window size and overlap for dividing text data. This method is discussed in detail in the subsequent section.
4. **Chunk the Text Data:** Split the documents into chunks using the identified optimal window size and overlapping strategy.
5. **Generate Embeddings:** Convert each text chunk into an embedding vector using the OpenAI Embedding technique. Store embeddings vector along with metadata in a ChromaDB vector database. The indexing structure is:

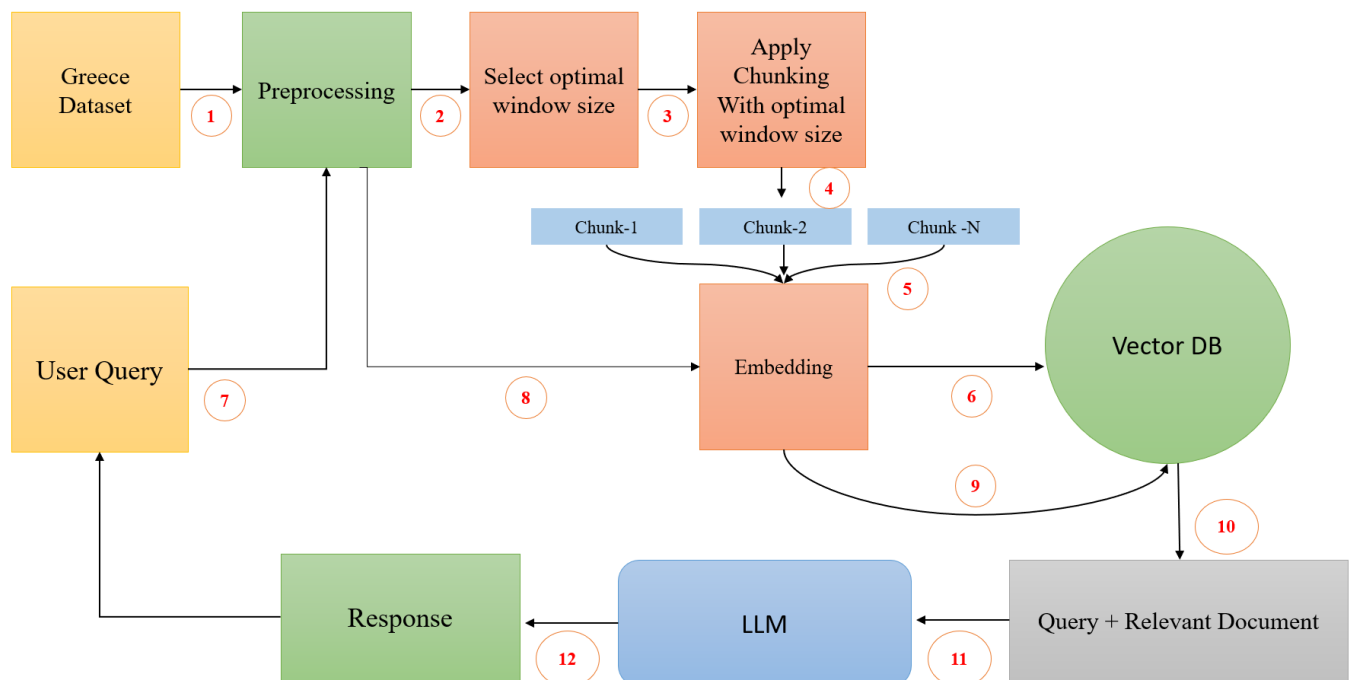
*{index\_no, chunk\_id, embedding\_vector, metadata}.*

6. **Process User Query:** Accept a user query and apply preprocessing step.
7. **Query Embedding:** Generate the embedding for the processed query using the OpenAI Embedding technique.
8. **Hybrid Retrieval and Re-ranking Technique:** Perform both keyword-based retrieval (BM25) and dense vector similarity search. Select the top 5 relevant documents and re-rank them using a cross-encoder transformer.
9. **Prepare LLM Input:** Combine the processed query and the top 5 relevant documents for LLM input.
10. **Configure LLMs and Prompts:** Define both open-source (LLaMA, Mistral) and closed-source (OpenAI) LLMs along with appropriate prompt templates and

parameters. The prompts are crafted to reduce hallucination and include source citations.

**11. Generate LLM Response:** Feed the processed query and relevant documents to the selected LLM. The model generates an answer along with source citations.

**12. Evaluate System Performance:** Evaluate the quality of the LLM-generated response and the retrieval system in the evaluation section.



**Figure 1:** RAG pipeline for Q & A Chatbot

## 2. DIVERSITY-BASED CHUNKING ALGORITHM ANALYSIS

This algorithm implements a simplified chunking optimization algorithm that identify optimal chunk size and overlap. The detailed steps are explained below:

### Step. 1: Apply preprocessing:

Load “Greece dataset” and apply preprocessing technique to all text file ranging from 1 to 60.

### Step. 2: Document chunking:

Let  $D$  is the document (processed text) is spitted into chunk  $C_i = \{C_1, C_2, \dots, C_N\}$ ,  $k$  is the different chunk size  $\{128, 256, \dots, 768\}$  and  $l=50$  (fixed overlap in token)

### Step. 3: Diversity Score Calculation:

Our aim is to calculate diversity score for each chunk. The diversity score says how semantically different two chunks from each other. It explains when you split a long text into smaller pieces, each piece covers a distinct idea or topic. The following steps to calculate diversity score:

#### a. Calculate embedding for each chunk:

$$E_i = \text{Embedding\_Model}(C_i)$$

#### b. Calculate pair wise similarity matrix:

Compute cosine similarity for all pair using below equation:

$$S_{i,j} = \text{Cosine\_Similarity}(E_i, E_j)$$

#### c. Calculate average similarity:

$$\dot{M} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N S_{i,j}$$

#### d. Calculate the diversity score:

$$D = 1 - \dot{M}$$

Higher Diverse score means more diverse chunks (lower average pairwise similarity).

#### e. Optimization Problem:

Our optimization problem is for each document  $D$ ; we will find the optimal chunk size that will maximize the diversity score means the optimal chunk has highest diverse score using following equation:

$$\dot{k} = \text{argmax}(D\{C_i(k, l)\})$$

Where:

$$k = \{128, 256, \dots, 768\} = \text{chunk sizes}$$

$$C_i(k, l) = \text{chunk generated with size } k \text{ and overlap } l$$

#### Step. 4: Statistical Analysis:

Across  $N$  documents, compute

**a. Optimal chunk sizes:**

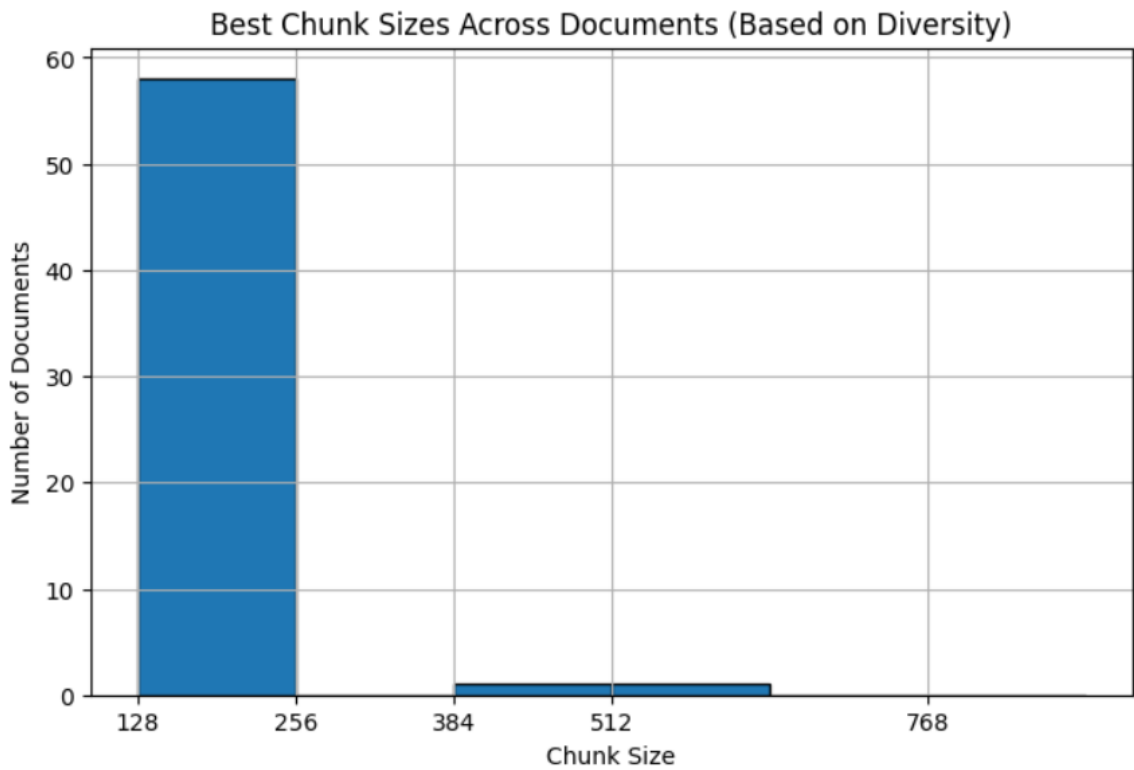
$$\hat{k} = \{k_1, k_2, \dots, k_N\}$$

**b. Histogram analysis:**

Plot empirical distribution of  $\hat{k}$  to identify optimal chunk sizes.

**Figure 2** shows the histogram distribution of the optimal chunk sizes selected for a set of documents, where "optimal" refers to the chunk size that maximizes semantic diversity (i.e., minimizes redundancy between chunks). The X-axis represents the candidate chunk sizes (128, 256, 384, 512, and 768 tokens), while the Y-axis indicates the number of documents for which a given chunk size was identified as the best. The height of each bar reflects how many documents performed best with the corresponding chunk size.

It is observed that the most frequently selected optimal chunk sizes are 128 and 256 tokens. However, considering both the higher diversity scores and the ability to capture longer contextual relationships, a chunk size of **256 tokens** is recommended.



**Figure 2:** Diversity-Based Chunking Algorithm

### 3. DOCUMENT CHUNKING

This section processes text files from a specified directory by dividing each document into smaller, overlapping segment referred to as *chunk*, using a recursive text splitting algorithm. The splitter is configured with a default chunk size of 256 tokens and an overlap of 50 tokens to preserve contextual continuity across chunks. This chunking strategy was applied to all 60 files in the dataset, resulting in a total of 516 chunks.

### 4. EMBEDDING TECHNIQUE AND STORING IN VECTOR DATA BASE

In this section, each chunk is converted into a vector representation using embedding techniques. Two embedding methods have been implemented: one open-source (SBERT) and the other closed source (OpenAI). The details of both embedding techniques are presented in **Table 1**. Due to its high dimensionality, rich semantic representation, and low latency, the OpenAI embedding technique is selected for further use.

**Table. 1:** Comparison between open source and closed source embedding model

| Feature               | Open source              | Closed Source                 |
|-----------------------|--------------------------|-------------------------------|
| Model Name            | SBERT (all-MiniLM-L6-v2) | OpenAI text-embedding-3-small |
| Dimension             | 384                      | 1536                          |
| Semantic relationship | Poor                     | Rich                          |
| Latency               | High                     | Low                           |

The chunks are converted into vectors using the OpenAI embedding technique and stored in a vector database. Among the various available vector databases, **ChromaDB** is used in this case due to its seamless compatibility with popular embedding models (e.g., OpenAI, Sentence Transformers, Cohere), its ability to run locally without requiring an external server or database, and its efficient in-memory retrieval powered by optimized approximate nearest neighbours (ANN) algorithms.

### 5. RETRIEVAL SYSTEM

This section is crucial for retrieving relevant documents based on the user's query. The quality of the LLM's response largely depends on how accurately relevant documents are retrieved from the vector database. If irrelevant documents are retrieved, the LLM's response may be poor. Conversely, retrieving contextually relevant documents leads to more accurate and informative answers. To improve the retrieval process, a **hybrid search with re-ranking approach** is adopted. The detailed steps for this technique are outlined below:

- a. **Query Embedding:** Convert user query into an embedding vector using OpenAI embedding technique.
- b. **Dense Vector Search:** Compute cosine similarity between the query embedding and document embeddings stored in the vector database. Retain documents with a similarity score above a predefined threshold (0.5).
- c. **BM-25 Keyword Search:** Perform a keyword-based search using the BM25 algorithm to retrieve documents containing query-relevant terms.
- d. **Merge Results:** Combine the results from both dense vector search and BM25 keyword search to create a candidate set of documents.
- e. **Semantic Filtering:** Use a Sentence Transformer (bi-encoder) to encode the query and all candidate documents. Filter out documents with semantic similarity scores below the threshold.
- f. **Re-ranking Technique using Cross-Encoder:** For each remaining (query, document) pair, use a cross-encoder model to assign a relevance score. Re-rank the documents based on these scores and return the top-k most relevant documents.

The main advantages of this technique are its ability to combine both sparse (lexical) and dense (semantic) search methods, the use of semantic filtering to ensure contextual relevance, and the application of a cross-encoder to rank documents based on relevance scores. This enables the system to effectively handle both keyword-based and meaning-based matches

## 6. LLM MODEL AND PROMPT ENGINEERING

This section outlines the models used in the chatbot system. Two categories of models were employed: open-source models (*LLaMA 3.2:1B*, *Gemma:2B*, and *Mistral: latest*) and OpenAI's model (*GPT-3.5-Turbo*). While open-source models were integrated for experimentation, it was observed that they had higher response latency and, in some cases, they failed to generate responses with proper source citation. As a result, *GPT-3.5-Turbo* was preferred due to its faster response time and consistent citation capabilities.

To ensure deterministic responses, the temperature parameter was set to  $T = 0.3$ . Higher temperature values tend to increase response creativity, while lower values promote consistency and determinism. After experimenting with values in the range  $T = \{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$ , it was found that  $T = 0.3$  yielded the most balanced results with accurate and cited responses.

To minimize hallucination in responses, prompt engineering techniques were employed. Since labelled data was not available, a zero-shot prompting approach was used. After experimenting with various prompt formulations, the following prompt was found to produce more accurate responses with reduced hallucination.

**Prompt:** {“You are a helpful assistant. Use the following sources to answer the question. Always cite the source in the format *[[SOURCE: name]]*. If the answer is not found, say: “I don't have enough information.”}

## 7. MODEL EVALUATION

In the absence of ground truth data, the effectiveness of the model is evaluated by analyzing the relationship between the query, the retrieved documents, and the LLM-generated response. The performance metrics used for this evaluation are listed in the **Table 2** below.

**Table. 2:** Model Evolution using various metrics.

| Metrics                                     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RAGAS</b>                                | Calculate <b>ragas score</b> , <b>faithfulness</b> and <b>answer_relevancy</b> using OpenAI API key.                                                                                                                                                                                                                                                                                                                            |
| <b>Detect Hallucination</b>                 | This metric evaluates whether the model's response contains hallucinations. <ol style="list-style-type: none"> <li>1. Compute the similarity score between the retrieved documents and the LLM response.</li> <li>2. If the similarity score is below the threshold (e.g., 0.5), the response is considered hallucinated; otherwise, it is not hallucinated.</li> </ol>                                                         |
| <b>ROUGE-L</b>                              | Calculate ROUGE-1, ROUGE-2 and ROUGE-L                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>BLEU</b>                                 | Calculate BLEU score                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Proxy Question Similarity Evaluation</b> | <ol style="list-style-type: none"> <li>1. Generate multiple proxy questions derived from the original query.</li> <li>2. Compute the similarity score between each proxy question and the LLM-generated response.</li> <li>3. Calculate the average of all similarity scores obtained.</li> <li>4. If the average similarity score exceeds a defined threshold (e.g., 0.5), the LLM response is considered relevant.</li> </ol> |

## 8. RETRIEVAL SYSTEM EVALUATION

Since ground truth data is not available, the quality of the retrieval system is evaluated by comparing the query with the retrieved documents. The following metrics, as shown in the **Table 3**, are used to assess the system's performance.

**Table. 3:** Evaluation of Retrieval system using various metrics.

| Metrics                            | Explanation                                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Semantic Similarity Scoring</b> | Calculate semantic similarity scoring for ranking retrieved documents based on their relevance to a user query using sentence embeddings.                                                         |
| <b>Keyword Coverage Scores</b>     | This metric evaluates how well retrieved documents contain the important keywords from a user query, providing a simple but effective measure of lexical matching quality.                        |
| <b>Content Density Scores</b>      | Measures the relevance of a document by calculating the density of query keywords, defined as the frequency of query terms appearing in the document divided by the total length of the document. |

## 9. RESULTS AND DISCUSSION

In order to evaluate the performance of the chatbot system in real-time, we provided two sets of questions. Set A contains questions related to the provided dataset, while Set B includes questions that are out of scope and not covered by the dataset. Sample questions from both sets are provided below:

### a. Set A-Question (Related to the Dataset):

- 1. What purpose did the tales of ancient Greek heroes serve beyond entertainment?*
- 2. How are Heracles's Twelve Labors interpreted symbolically?*
- 3. Who were the main leaders of the Persian Empire during the Persian Wars?*
- 4. Which Persian ruler led the full-scale invasion of Greece in 490 BCE?*
- 5. What is one of the most enduring political contributions of Ancient Greece?*
- 6. What were some areas that Aristotle contributed to?*
- 7. Which Ancient Greek philosopher introduced the concept of "ideal forms"?*
- 8. Which year the Alexander was born?*

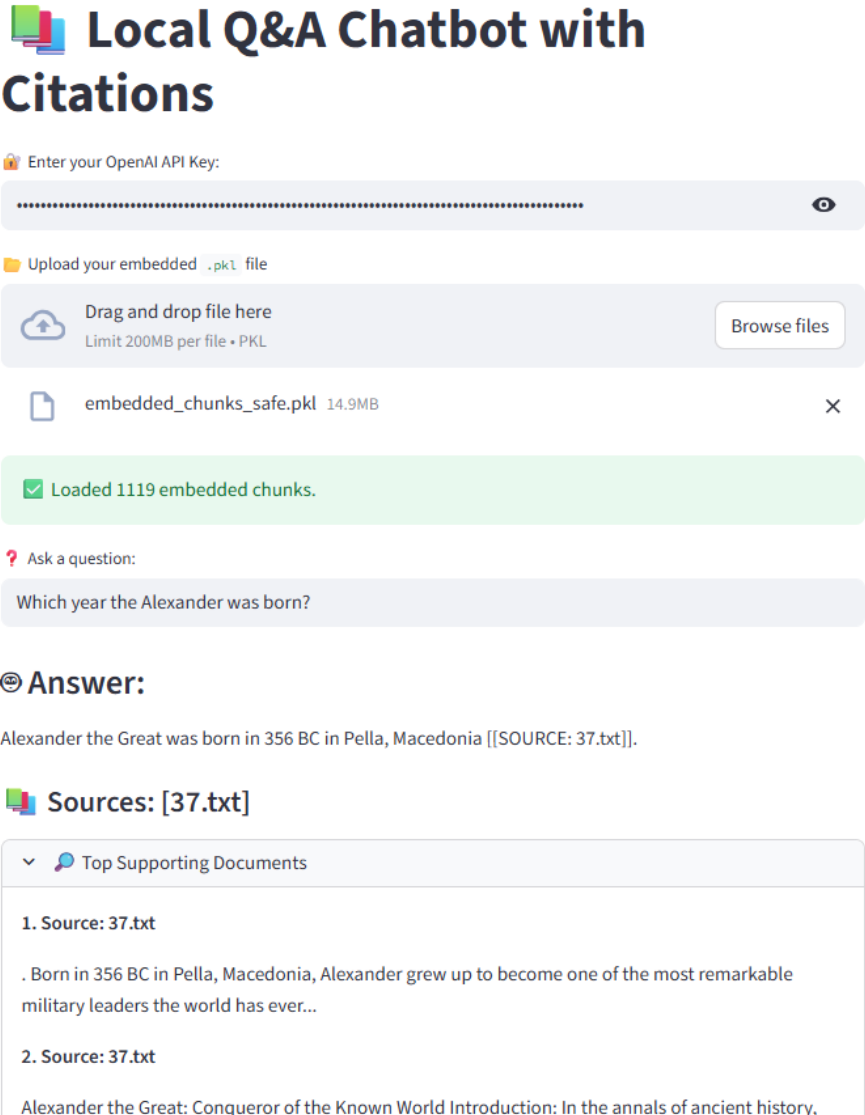
### b. Set B-Questions (Not Related to the Dataset):

- 1. Explain the attention mechanism in transformers?*

2. *What are the key differences between generative AI and traditional AI?*
3. *What are foundation models, and why are they important in modern AI development?*
4. *Who is the CEO of SpaceX?*
5. *What is the current GDP of India?*
6. *How is AI being applied in financial fraud detection and risk analysis?*
7. *What are the advantages and risks of using Retrieval-Augmented Generation (RAG) in production systems?*

**c. Output of Set A-Questions:**

All questions from **Set A** were passed to the LLM model, and it was observed that the retrieval pipeline effectively extracted documents relevant to the queries. These relevant documents were then provided to the LLM model, which generated accurate responses with appropriate source citations and no hallucinations. Screenshots of a few sample results are attached below.



## Local Q&A Chatbot with Citations

Enter your OpenAI API Key:

Upload your embedded .pkl file

Drag and drop file here  
Limit 200MB per file • PKL

embedded\_chunks\_safe.pkl 14.9MB

Loaded 1119 embedded chunks.

Ask a question:

Which year the Alexander was born?

**Answer:**

Alexander the Great was born in 356 BC in Pella, Macedonia [[SOURCE: 37.txt]].

**Sources: [37.txt]**

Top Supporting Documents

- Source: 37.txt**  
. Born in 356 BC in Pella, Macedonia, Alexander grew up to become one of the most remarkable military leaders the world has ever...
- Source: 37.txt**  
Alexander the Great: Conqueror of the Known World Introduction: In the annals of ancient history,

# Local Q&A Chatbot with RAGAS Evaluation

Enter your OpenAI API Key:

.....

Upload your embedded .pkl file



Drag and drop file here

Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✓ Loaded 1119 embedded chunks.

Ask a question:

What purpose did the tales of ancient Greek heroes serve beyond entertainment?

## Answer:

The tales of ancient Greek heroes served beyond entertainment by also serving as important cultural symbols, explaining natural phenomena and human experiences, educating the ancient Greeks, providing them with a moral compass, and inspiring and guiding individuals as cautionary tales or examples of virtue [[SOURCE: 8.txt]], [[SOURCE: 9.txt]], [[SOURCE: 5.txt]], [[SOURCE: 7.txt]].

Sources: [5.txt], [9.txt], [8.txt], [7.txt]

# Local Q&A Chatbot with RAGAS Evaluation

Enter your OpenAI API Key:

.....

Upload your embedded .pkl file



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Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✓ Loaded 1119 embedded chunks.

Ask a question:

How are Heracles's Twelve Labors interpreted symbolically?

## Answer:

The interpretation of Heracles's Twelve Labors as symbolic representations of the challenges and struggles faced by him can be found in various sources. The Twelve Labors were tasks imposed upon Heracles by King Eurystheus as a form of atonement for the murder of his wife [[SOURCE: 7.txt]]. These labors are often viewed as symbolic representations of the hero's journey, personal growth, and overcoming obstacles [[SOURCE: 9.txt]].

Sources: [7.txt], [9.txt]

The model metrics for each question are calculated and presented in the table. The table includes **RAGAS** metrics (**ragas score, faithfulness, answer relevance**), **hallucination status, proxy question similarity score**, ROUGE score, and BLEU score. Retrieval system metrics are not included in the table since ground truth data is not available. However, these metrics can be observed in the Jupyter Notebook. The result of 5 questions is shown in the **Table 4**.

**Table. 4:** Evaluation of the model using few samples dataset.

| Question No | Ragas score | Faithfulness | Answer relevance | Hallucination | Proxy question similarity score |
|-------------|-------------|--------------|------------------|---------------|---------------------------------|
| 1           | 0.95        | 1.0          | 0.92             | No            | 0.84                            |
| 2           | 0.65        | 0.50         | 0.96             | No            | 0.83                            |
| 3           | 1.0         | 1.0          | 1.0              | No            | 0.82                            |
| 4           | 0.97        | 1.0          | 0.95             | No            | 0.86                            |
| 5           | 0.97        | 1.0          | 0.95             | No            | 0.65                            |

#### d. Output of Set B-Questions:

All questions from **Set B** were passed to the LLM model, and it was observed that the retrieval pipeline could not extract any relevant documents, as these queries were outside the scope of the dataset. As a result, the LLM did not generate any meaningful response. Screenshots of a few sample queries from **Set B** are provided below, clearly showing that since the queries are unrelated to the dataset, the system returns the response: *"No Relevant Documents Found."*. Since no relevant documents were found, the model metrics could not be calculated.

# Local Q&A Chatbot with RAGAS Evaluation

🔑 Enter your OpenAI API Key:

.....



📁 Upload your embedded .pkl file



Drag and drop file here

Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✅ Loaded 1119 embedded chunks.

? Ask a question:

What are the key differences between generative AI and traditional AI?

⚠️ No relevant documents found.

# Local Q&A Chatbot with RAGAS Evaluation

🔑 Enter your OpenAI API Key:

.....



📁 Upload your embedded .pkl file



Drag and drop file here

Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✅ Loaded 1119 embedded chunks.

? Ask a question:

What are foundation models, and why are they important in modern AI development?

⚠️ No relevant documents found.



# Local Q&A Chatbot with RAGAS Evaluation

🔑 Enter your OpenAI API Key:

.....



📁 Upload your embedded .pkl file



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Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✅ Loaded 1119 embedded chunks.

? Ask a question:

Who is the CEO of SpaceX?

⚠️ No relevant documents found.



# Local Q&A Chatbot with RAGAS Evaluation ↻

🔑 Enter your OpenAI API Key:

.....



📁 Upload your embedded .pkl file



Drag and drop file here

Limit 200MB per file • PKL

Browse files



embedded\_chunks\_safe.pkl 14.9MB



✅ Loaded 1119 embedded chunks.

? Ask a question:

How is AI being applied in financial fraud detection and risk analysis?

⚠️ No relevant documents found.

## 10. CHALLENGES IN THE CURRENT CHATBOT

There are many challenges in the current chatbot system:

- a. **Latency issue:** Due to hardware constraints (limited to 16 GB RAM), open-source LLMs (**llama3.2:1b**, **gemma:2b** and **mistral: latest**) exhibit high response times (latency). To mitigate this, the system currently relies on OpenAI's API, which offers faster inference but introduces dependency on external services.
- b. **Single conversational chatbot:** The current chatbot lacks memory capabilities, restricting it to single-turn conversations without retaining chat history. This limitation stems from both memory constraints and the cost implications of storing and retrieving conversation logs via API.
- c. **Limited evaluation metrics:** The absence of ground truth data in the dataset restricts comprehensive evaluation. Without benchmark question-answer pairs, only basic metrics can assess response quality, limiting insights into model accuracy and relevance. For example, the current chatbot system is evaluated using **RAGAS**. However, due to the lack of ground truth data, only metrics such as **ragas score**, **faithfulness** and **answer\_relevancy** are applied, as these do not require ground truth. Other **RAGAS** metrics like **context precision** and **context recall** are excluded since they depend on ground truth for assessment.
- d. **Problem of Diversity-Based Chunking Algorithm:** The current Diversity-Based Chunking Algorithm uses a fixed overlap of ( $l=50$ ) with chunk size of  $k = \{128, 256, \dots, 768\}$ . However, testing dynamic overlaps  $l=\{50, 100, 150, 200, \dots\}$  could improve the diversity score with optimal window size and overlap chunk.
- e. **Zero-Shot Prompt Technique:** Due to the absence of labelled data, zero-shot prompting was explored. However, to further enhance the model's performance, few-shot prompting or chain-of-thought prompting can be employed.

## 11. FUTURE WORK

To address the limitations of the current chatbot system, the following enhancements are recommended for future implementation:

- a. **Incorporate Ground Truth Evaluation:** In order to assess the quality of both the LLM responses and the retrieval system, ground truth data should be added to the dataset. This involves generating question-answer pairs to evaluate the accuracy and relevance of responses, enabling systematic benchmarking of model performance.

- b. **Implement Chatbot with Memory:** The existing chatbot lacks chat history retention due to the absence of memory functionality. Future iterations should integrate a memory mechanism or employ an AI agent capable of storing and referencing past interactions to improve contextual understanding and coherence.
- c. **Adopt Agentic RAG:** To optimize retrieval and generation, an Agentic RAG framework can be implemented. This approach utilizes multiple AI agents to refine user queries, analyse retrieved documents, and dynamically invoke APIs or tools, thereby improving response accuracy and system efficiency.
- d. **Implement HyDE:** The Hypothetical Document Embedding (HyDE) technique can enhance retrieval performance by generating a hypothetical answer to a query, embedding it, and using it for semantic similarity searches in the vector database. By combining the original query, hypothetical answer, and retrieved documents, the LLM can produce more accurate responses with reduced hallucination.
- e. **Optimize Diversity-Based Chunking Algorithm:** The current chunking algorithm has limitations in determining optimal chunk sizes and overlaps. A dynamic chunking approach should be explored, where varying overlap chunk sizes  $\{50, 100, 150, 200...\}$  tokens are tested to identify the most optimal chunk size and overlap chunk for retrieval accuracy and computational efficiency.
- f. **Apply Fine Tuning Technique:** To further enhance chatbot performance, fine-tuning a pre-trained LLM on a custom dataset is recommended. Expanding the dataset with additional high-quality question-answer pairs will improve model adaptation. Efficient fine-tuning methods such as LoRA (Low-Rank Adaptation) or QLoRA (Quantized LoRA) can be employed to optimize performance while minimizing computational overhead.
- g. **Better Prompting Technique:** To improve the chatbot's performance, more effective prompting strategies such as few-shot prompting can be utilized. By generating high-quality question-answer pairs and incorporating them as examples in the prompt, the model can better understand the context and deliver more accurate and relevant responses.
- h. **Use LLM as Judge:** To assess the performance of the chatbot, an additional LLM can be employed to evaluate its responses. However, this approach increases both computational cost and response latency due to multiple LLM calls.

## **12. CONCLUSION**

The RAG pipeline developed for the Q&A chatbot effectively answers queries related to the provided dataset, including proper source citations. However, it correctly refrains from responding to questions outside the dataset's scope. This demonstrates that the chatbot is robust, with minimal hallucination.